

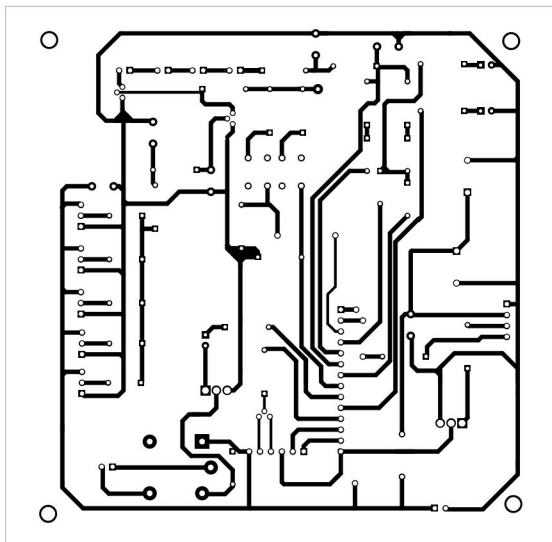


Fig. 2: Actual-size PCB pattern of the Arduino based GSM home security system

and LED1 lights up to indicate that the system has been armed. Any unintentional intrusion by you at this point will trigger an alert. So be careful. Even otherwise, you will have 30 seconds to leave home and close the doors.

As shown in Fig. 1, T3 has been used to turn on GSM module via Arduino software. So there is no need to manually start GSM shield. RF chokes (RFCs) L1 and L2, along with decoupling capacitors C3 through C6, filter out stray radio frequencies and noise to avoid false triggering of the alarm system.

LED3 lights up when the alarm is activated. Board 1's pin 6 (D6) has been used to activate relay RL1. A loud siren can be switched on via the relay to scare the burglar or intruder away. Reed switches

can be mounted on door frames of entry and exit doors. Small magnets should be mounted on the doors themselves. When a door is closed, the magnet and the reed switch should be in close contact with each other. PIR sensors should be placed in each room, preferably on the wall and in view of the window.

Pressing switch S2 deactivates the alarm. The switch can be placed in a hidden location

outside the house, but it is not recommended. Leave the house after pressing switch S1 and turn off the alarm via SMS when coming home from work in the evening.

Construction and testing

An actual-size, single-side PCB pattern for the Arduino based GSM home security system is shown in Fig. 2 and its component layout in Fig. 3. The circuit can be assembled on the PCB. It can fit easily inside most prototype/custom enclosures.

Enter your primary mobile phone number to receive calls and SMS. This number is also used to send SMS commands like '\$ON' and '\$OFF1234'. Replace xxxxxxxxxx with your ten-digit mobile number in the following line of Arduino1.ino sketch:

```
String number = "xxxxxxxxxx";  
// 10 digit mobile number
```

Enter the second number to receive alerts (phone calls only) in all the following lines under void call():

```
mySerial.print("ATDT+91xxxxxxxxxx");
```

As shown in Fig. 1, Arduino Board 1 and GSM module get ground connection through transistor TIP142 (T1). Also, very little drive current (<5mA) is required from Board 1. A 12V, 2A regulated power supply is recommended. It should be able to deliver 2A of current for the GSM module. An optional battery backup system can be added for additional security in case the burglar decides to turn off the electricity.

House the circuit in a plastic box and place it in a locked wooden cabinet. Make sure the GSM module has good signal strength. The alarm system can also be used to protect

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com